

Generation Interconnection System Impact Study Report

For Queue Project Q20-1558

Sunset Road 138 kV

March 31, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1558. The Interconnection Customer proposes a 70.3 MW battery energy storage facility interconnecting at the Sunset Road 138 kV substation in the Virginia area. The request was received on August 25, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 475 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,428,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kiwi / North Lake 138 KV line	Easton 2 / Vienna Operations 345 KV	113.1	528	24.7
Jarratt / Sandburg 138 KV	Vienna Operations / Easton2 345 KV Line	111.6	542	27.1
Sherwood / Sandburg 138 KV	Eastn 2 / Vienna Operations 345 KV	111.2	554	27.1
Lammers / Chemolite 138 KV line	Vienna Operations / Easton 2 345 KV	111.2	318	15.3
Hitchland / Sherwood 138 KV line	Vienna Operations / Easton 2 345 KV	111.0	561	27.1
Hitchland / N Lake 138 KV	Vienna Operations Substation / Easton 2 345 KV	110.0	591	27.1
Wye Mills / Holston Dam 138 KV	Cady / Easton 2 345 KV	106.0	231	6.5
W -Station Substation / St Bonifacius 138 KV	Southtown / Vienna Operations 345 KV	104.6	1,362	54.6
Newmrk Substation / Ess Lewes 138 KV	Vienna Operations / Easton 2 345 KV	104.0	427	25.0
Ranall County / Chemolite 138 KV	Vienna Operations / Easton 2 345 KV	103.1	330	15.3
Park Ave / Hillsboro 138 KV line	Cady / Easton 2 345 KV	101.6	760	24.9
Holston Dam / Carancahua 138 KV	Cady / Easton 2 345 KV	100.3	235	6.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kiwi / North Lake 138 KV line	Rebuild line with higher-capacity conductor	\$6,258,000	2% (\$153,000)
Jarratt / Sandburg 138 KV	Rebuild line with higher-capacity conductor	\$12,304,000	2% (\$296,000)
Sherwood / Sandburg 138 KV	Rebuild line with higher-capacity conductor	\$5,247,000	2% (\$126,000)
Chemolite / Lammers 138 KV line	Rebuild line with higher-capacity conductor	\$6,581,000	2% (\$160,000)
Hitchland / Sherwood 138 KV line	Rebuild line with higher-capacity conductor	\$7,921,000	2% (\$191,000)
N Lake Substation / Hitchland 138 KV L	Reconductor line	\$3,929,000	2% (\$95,000)
Wye Mills / Holston Dam 138 KV	Reconductor line	\$3,278,000	2% (\$70,000)
St Bonifacius / West -Station 138 KV	Reconductor line	\$2,700,000	2% (\$65,000)
Newmark / Ess Lewes Substation 138 KV	Reconductor line	\$3,963,000	2% (\$99,000)
Cheolite / Randall County Substation 138 KV	Reconductor line	\$3,123,000	2% (\$76,000)
Hillsboro / Park Ave 138 KV	Reconductor line	\$1,527,000	2% (\$31,000)
Holston Dam / Carancahua 138 KV	Reconductor line	\$3,090,000	2% (\$66,000)
	Total allocated to project		\$1,428,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.